

Thermodynamics of the Quantum Mpemba Effect

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We investigate the quantum Mpemba effect from the perspective of nonequilibrium quantum thermodynamics by studying relaxation dynamics of quantum systems coupled to a Markovian heat bath, which are described by Davies maps. Starting from a state with coherences in the energy eigenbasis, we demonstrate that an exponential speedup to equilibrium will always occur if the state is transformed to a diagonal state in the energy eigenbasis, provided that the spectral gap of the generator is defined by a complex eigenvalue. When the transformed state has a higher nonequilibrium free energy, we argue using thermodynamic reasoning that this is a *genuine* quantum Mpemba effect. Furthermore, we show how a unitary transformation on an initial state can always be constructed to yield the effect and demonstrate our findings by studying the dynamics of both the nonequilibrium free energy and the irreversible entropy production in single and multiqubit examples.

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The Mpemba effect describes a situation where an initially hot system is quenched into a cold bath and reaches equilibrium faster than an initially cooler system. Although the first systematic investigations of this phenomenon were performed by Mpemba, Osborne, and Kell [1,2] in the late sixties, it had been discussed by Aristotle over 2000 years ago [3] and noticed by others, such as Descartes [4] and Bacon [5] throughout history. Since Mpemba and Osborne's original work, the effect has been explored in an increasingly diverse range of physical systems such as clathrate hydrates [6], polymers [7], magnetic alloys [8], carbon nanotube resonators [9], granular gases [10], and dilute atomic gases [11]. However, despite the wide range of studies, the physical origin and even the very existence of the phenomenon are still debated in the literature [12,13].

A breakthrough in the understanding of this anomalous effect came from analysing the stochastic thermodynamics of Markovian systems [14,15]. At long times, the dynamical evolution of the state is a linear combination of the stationary state and the system's slowest decaying mode. A Mpemba effect can occur if the initial state with higher temperature has a smaller amplitude with this mode, with an exponential speedup to equilibrium occurring if the

amplitude goes to zero (known as the strong Mpemba effect) [15]. This framework has been used recently to experimentally probe the Mpemba effect [16] and its inverse [17] under controlled conditions and to boost thermalization protocols [18].

Recently work has been undertaken to generalize the Mpemba effect to quantum dynamics. The phenomenon and the framework generalize naturally to open quantum systems [19–29] and this has lately been explored experimentally in ion traps [30,31]. However, it should be stated that very few of these studies focus specifically on thermalization but rather more generally on anomalous relaxation to general fixed points. Another series of works have investigated a quantum Mpemba-like effect in isolated systems, related to symmetry restoration in quenched dynamics starting from states that break the symmetry of the generating Hamiltonian [32–36].

In this Letter, we focus specifically on Davies maps which are an important class of continuous-time quantum dynamical semigroups describing thermalization. These maps rigorously describe the thermalization of a quantum system when weakly coupled to a heat bath [37–39] and are often said to be the quantum equivalent of classical Glauber dynamics. The special mathematical properties of these maps allow us to identify a unique quantum Mpemba effect that stems from the subblock structure of the generator in the energy eigenbasis. Starting from a state that has coherences, we show how a unitary transformation can transform the state to generate an exponential speed-up which will always occur when the spectral gap of the

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generator is defined by a complex eigenpair. When the transformed state has a higher nonequilibrium free energy [40–42], the dynamics give rise to a genuine quantum Mpemba effect. Furthermore, we study the division of the irreversible entropy production [43,44] into coherent and incoherent parts [45,46]. Our findings are illustrated on single-qubit and many-qubit examples.

Consider a quantum Markovian master equation that evolves a density matrix $\hat{\rho}(t)$

$$\frac{d\hat{\rho}(t)}{dt} = \mathcal{G}\hat{\rho}(t). \quad (1)$$

Here $\mathcal{G}$ is the generator, known as the Lindbladian, consisting of a unitary part and a dissipative part such that $\mathcal{G} = i\delta + \mathcal{D}$ with $\delta(\cdot) = -[\hat{H}, \cdot]$, $\hat{H}$ being the system Hamiltonian and $\mathcal{D}(\cdot) = \sum_l \hat{L}_l(\cdot)\hat{L}_l^\dagger - \frac{1}{2}\{\hat{L}_l^\dagger\hat{L}_l, (\cdot)\}$, where $\hat{L}_k$ are the so-called jump operators. Using the generator we can write down the general evolution of an initial density matrix $\hat{\rho}_i$ as

$$\hat{\rho}(t) = e^{\mathcal{G}t}\hat{\rho}_i = \hat{\tau} + \sum_{k=2}^{D^2} \text{Tr}(\hat{l}_k \hat{\rho}_i) \hat{r}_k e^{\lambda_k t}, \quad (2)$$

where $D = \dim(\hat{H})$ and we assume that $\hat{\tau}$ is the unique steady state that is the right eigenoperator of the generator $\mathcal{G}$ associated with the zero eigenvalue. Here $\hat{l}_k$ and $\hat{r}_k$ are the left and right eigenoperators corresponding to the eigenvalue λ_k such that $\mathcal{G}[\hat{r}_k] = \lambda_k \hat{r}_k$ and $\mathcal{G}^\dagger[\hat{l}_k] = \lambda_k \hat{l}_k$, where $\mathcal{G}^\dagger$ is the adjoint generator that acts on observables rather than states. The superoperator $\mathcal{G}$ preserves the Hermiticity of $\hat{\rho}(t)$, which implies that if λ_k is a complex eigenvalue then λ_k^* is also. We order the eigenvalues in ascending order according to the modulus of their real part such that $0 = \lambda_1 < |\text{Re}(\lambda_2)| \leq |\text{Re}(\lambda_3)| \leq \dots$. The spectral gap is then $|\text{Re}(\lambda_2)|$ and defines the longest timescale in the system such that $|\hat{\rho}(t) - \hat{\tau}| \propto \exp[\text{Re}(\lambda_2)t]$. In [20] the authors show that an exponential speed up to the steady state can be achieved by finding a unitary that acts on the initial state such that

$$\text{Tr}(\hat{l}_2 \hat{U} \hat{\rho}_i \hat{U}^\dagger) = 0, \quad (3)$$

so that $|\hat{\rho}(t) - \hat{\tau}| \propto \exp[\text{Re}(\lambda_3)t]$. The key assumptions in [20] were a pure initial state and a spectral gap defined by a real eigenvalue. When the lowest eigenmodes of the generator form a complex conjugate pair at long times one has

$$\hat{\rho}(t) \propto \hat{\tau} + e^{\text{Re}(\lambda_2)t} \left[\text{Tr}(\hat{l}_2 \hat{\rho}_i) \hat{r}_2 e^{i\text{Im}(\lambda_2)t} + \text{Tr}(\hat{l}_2^\dagger \hat{\rho}_i) \hat{r}_2^\dagger e^{-i\text{Im}(\lambda_2)t} \right], \quad (4)$$

so for a strong Mpemba effect, one would need to find a unitary such that both $\text{Tr}(\hat{l}_2^\dagger \hat{U} \hat{\rho}_i \hat{U}^\dagger) = 0$ and

$\text{Tr}(\hat{l}_2 \hat{U} \hat{\rho}_i \hat{U}^\dagger) = 0$ [23]. As argued in [23], $\hat{l}_2$ is non-Hermitian so the strong Mpemba effect cannot be found using the same logic as in [20]. Here we demonstrate an alternative route to obtain such a unitary which exploits the mathematical structure of the generator in thermalizing open systems.

We consider Davies maps, which are defined by two properties. First, the unitary and dissipative part of the generator commute, and second they obey quantum detailed balance [47] with respect to the thermal state $\hat{\tau}_\beta = e^{-\beta \hat{H}}/Z$, with $Z = \text{Tr}[\exp(-\beta \hat{H})]$. Mathematically quantum detailed balance means that given two arbitrary operators $\hat{A}$ and $\hat{B}$, the unitary and dissipative parts of the generator obey $\langle \hat{A}, \mathcal{D}^\dagger(\hat{B}) \rangle_{\hat{\tau}_\beta} = \langle \mathcal{D}^\dagger(\hat{A}), \hat{B} \rangle_{\hat{\tau}_\beta}$ and $[\hat{H}, \hat{\tau}_\beta] = 0$, where $\langle \hat{A}, \hat{B} \rangle_{\hat{\tau}_\beta} = \text{Tr}(\hat{\tau}_\beta \hat{A}^\dagger \hat{B})$ is the Kubo-Martin-Schwinger (KMS) inner product. The eigenmatrices of the unitary part δ are the elemental matrix units $|n\rangle\langle m|$ with eigenvalues $\omega_{nm} = E_n - E_m$. Since this is a Davies map, the dissipative part $\mathcal{D}$ shares a common eigenspace with δ . Furthermore, if the Hamiltonian is nondegenerate, then the overall Davies map $e^{\mathcal{G}t}$ is block diagonal in the energy eigenbasis. We can write $\mathcal{G} = \mathcal{G}_P \oplus \mathcal{G}_C$, where $\mathcal{G}_P = \mathcal{D}_P$ is the population sub-block, whose right eigenmatrices are diagonal in the energy eigenbasis $|n\rangle\langle n|$ and $\mathcal{G}_C = i\delta + \mathcal{D}_C$ is the coherence sub-block. Note that due to the detailed balance condition the eigenvalues corresponding to the off-diagonal eigenmatrices of $\mathcal{D}_C$ are real whereas imaginary eigenvalues are due to the unitary part δ . The slowest decaying modes shown in Eq. (4) and indeed more generally $\hat{r}_i$ for $i \geq 2$ will be off-diagonal matrices in the energy eigenbasis. This ensures that they are traceless, which guarantees that the dynamics preserve normalization at all times. However, not only does δ commute with $\mathcal{D}$, it also commutes with $\mathcal{D}^\dagger$. This means that the left eigenvectors of $\mathcal{G}_C$ are also purely off-diagonal matrices in the energy eigenbasis since $\mathcal{G}^\dagger[\hat{l}_k] = \lambda_k \hat{l}_k$.

We now state our first finding: given an initial state $\hat{\rho}_i$ which has coherences in the energy eigenbasis, under an evolution with a Davies generator which has a spectral gap defined by the real part of a complex eigenpair, an exponential speed up towards the fixed point can always be achieved if a unitary $\hat{U}$ is performed to bring the state to be diagonal in the energy eigenbasis. Since all $\hat{l}_m$ with $\lambda_m \in \mathbb{C}$ are purely off-diagonal matrices in the energy eigenbasis then all associated amplitudes will be simultaneously eliminated $\text{Tr}(\hat{l}_m \hat{U} \hat{\rho}_i \hat{U}^\dagger) = 0$. We note that alternative protocols exploiting quantum coherences to speed up equilibration have been proposed in [48,49].

We want to stress an essential point: an exponential speedup in itself does not constitute a *genuine* Mpemba effect. In the original breakthrough paper [14], which focuses on thermodynamics, the effect is defined according to states with three temperatures $T_b < T_c < T_h$ which pertain to the fixed point, colder and hotter initial states, respectively. A Mpemba effect occurs if some time t_m exists

after which $D[\hat{\rho}^h(t), T_b] < D[\hat{\rho}^c(t), T_b]$. Here $D[\hat{\rho}(t), T]$ denotes a distance measure between the time-evolving state and the fixed point. Since we are dealing with nonequilibrium quantum states with possible coherences in the energy eigenbasis we do not have any initial notion of temperature. We have one point of equilibrium, namely, the fixed point $\hat{\tau}_\beta$. For quantum dynamics, we propose to use the nonequilibrium free energy, which is defined as

$$F_{\text{neq}}[\hat{\rho}(t)] = \text{Tr}[\hat{H}\hat{\rho}(t)] + \frac{1}{\beta} \text{Tr}[\hat{\rho}(t) \ln \hat{\rho}(t)]. \quad (5)$$

We consider the initial scenario $F_{\text{neq}}[\hat{\rho}'(t_0)] > F_{\text{neq}}[\hat{\rho}(t_0)] > F_{\text{eq}}$ where $\hat{\rho}'(t_0) = \hat{U}\hat{\rho}(t_0)\hat{U}^\dagger$ is the state following unitary transformation and $F_{\text{eq}} = -\beta^{-1} \ln Z$ is the equilibrium free energy of the fixed point. A quantum Mpemba effect occurs if some time t_m exists such that $F_{\text{neq}}[\hat{\rho}'(t)] < F_{\text{neq}}[\hat{\rho}(t)]$ for all times $t > t_m$. We call this a *genuine* quantum Mpemba effect to differentiate it from exponential speedups achieved by other transformations. We emphasize that for the speedup to be a quantum Mpemba effect the nonequilibrium free energy curves need to cross in time. The nonequilibrium free energy has several additional features that make it ideal to define and analyze the quantum Mpemba effect. First of all, it can be written as

$$F_{\text{neq}}[\hat{\rho}(t)] = \beta^{-1} D[\hat{\rho}(t) || \hat{\tau}_\beta] + F_{\text{eq}}(\hat{\tau}_\beta), \quad (6)$$

where $D(\hat{\rho} || \hat{\sigma}) = \text{Tr}[\hat{\rho}(\ln \hat{\rho} - \ln \hat{\sigma})]$ is the quantum relative entropy. The Klein inequality guarantees the positivity of the relative entropy, ensuring that $F_{\text{neq}}[\hat{\rho}(t)] \geq F_{\text{eq}}(\hat{\tau}_\beta) \forall t$. The quantum relative entropy is a very stringent measure of the distinguishability of two quantum states. While not itself a metric, it still upper bounds the trace distance via Pinsker's inequality $D(\hat{\rho} || \hat{\sigma}) \geq \|\hat{\rho} - \hat{\sigma}\|_1^2/2$, which captures the optimal distinguishability of quantum states with a single measurement. In addition, Eq. (5) gives us a prescription to find operations that transform an initial state to a state that has both a higher F_{neq} and also is diagonal in the energy eigenbasis. A simple operation that does this is a unitary which diagonalizes the state in the energy eigenbasis and then does a population inversion (see Supplemental Material [50]). When the generator has a spectral gap defined by a complex eigenpair this will *always* yield a quantum Mpemba effect since all overlaps with coherent modes go to zero. Finally, not only is the nonequilibrium free energy directly connected to the Spohn entropy production rate [43] as $\Pi = -\beta^{-1} dF_{\text{neq}}[\hat{\rho}(t)]/dt$, but as shown in [45,46] and in the Supplemental Material [50], the total irreversible entropy produced can be divided as

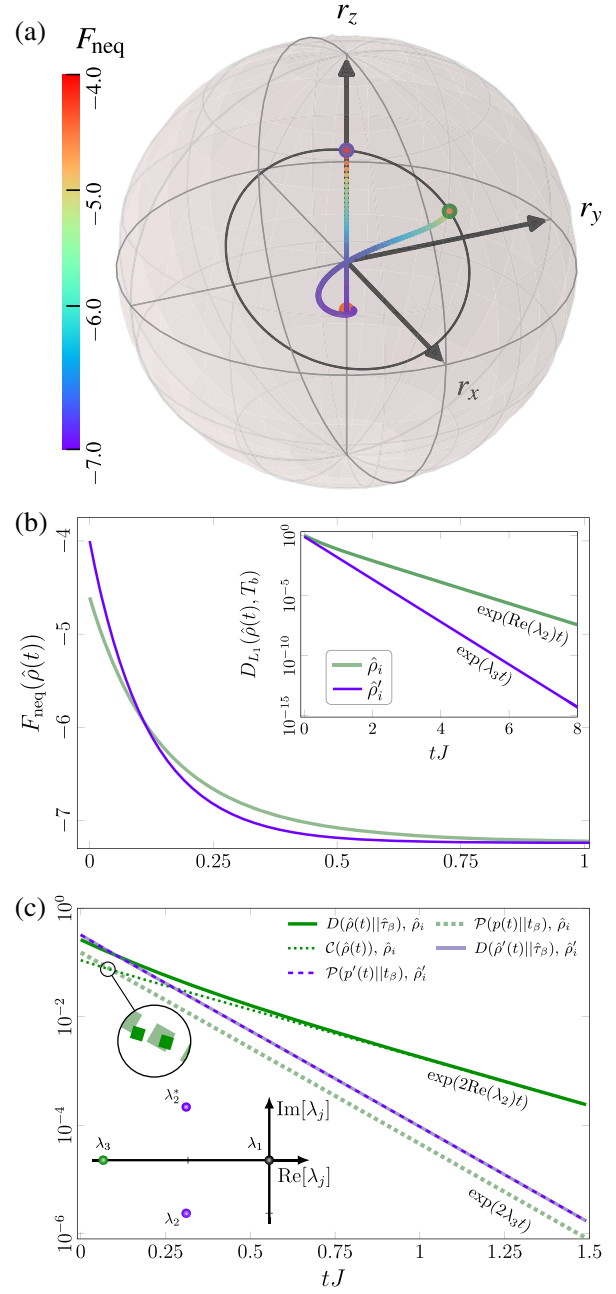


FIG. 1. The genuine quantum Mpemba effect in a single-qubit system. (a) Bloch sphere representation of the equilibration of a random state $\hat{\rho}_i$ (green dot) and the transformed state $\hat{\rho}'_i = \hat{U}\hat{\rho}_i\hat{U}^\dagger$ (purple dot) towards the thermal steady state (orange dot). The nonequilibrium free energy F_{neq} for the two thermalization processes is represented by the color scheme. (b) F_{neq} is initially higher for the optimized state (purple line), but drops below the random state's one (green line) at a later time. This crossing shows the occurrence of a genuine quantum Mpemba effect. Inset: The L_1 distances from the steady state indicate that $\hat{\rho}'_i$ thermalizes exponentially faster than $\hat{\rho}_i$. (c) Dynamics of the total entropy production D and its classical $\mathcal{P}$ and coherent $\mathcal{C}$ contributions. The rotation $\hat{U}$ yields an exponential speed up. Inset: spectrum of the Davies generator $\mathcal{G}$ for a single qubit. Here $\omega = 5J$, $\mathbf{r}_i = (0.276, 0.359, 0.303)$, $\mathbf{r}'_i = (0., 0., 0.545)$, and $T_b = 10J$.

$$D[\hat{\rho}(t)||\hat{\tau}_\beta] = \mathcal{P}[p(t)||\hat{\tau}_\beta] + \mathcal{C}[\hat{\rho}(t)], \quad (7)$$

where $\mathcal{P}[p(t)||t_\beta] = \sum_n p_n(t) \ln[p_n(t)/t_n^\beta]$ is the classical relative entropy of the time-evolving population vector and the thermal state and $\mathcal{C}[\hat{\rho}(t)] = S\{\Delta[\hat{\rho}(t)]\} - S[\hat{\rho}(t)]$ is known as the relative entropy of coherence where $\Delta[\hat{\rho}_s(t)]$ is the state dephased in the systems' energy eigenbasis. This is known to be an operational measure of quantum coherence [62]. In the following, we consider the thermalization of single and multiqubit systems. The characteristic energy scale is J and we set $k_B = 1$ and the dissipation strength $\gamma = J$ (see Supplemental Material [50] for details).

Example 1: Single qubit—The qubit is prepared in a random state $\hat{\rho}_i = 1/2(\mathbb{1} + \mathbf{r} \cdot \hat{\boldsymbol{\sigma}})$, where $\mathbf{r}$ is the Bloch vector and $\hat{\boldsymbol{\sigma}}$ is the vector containing the three Pauli matrices. The thermalization is described by a unitary part governed by the Hamiltonian $\hat{H} = \omega/2\hat{\sigma}_z$ and a dissipative part $\mathcal{D}$ defined according to the Davies map for a bosonic particle and a bath at temperature T_b . We apply a unitary $\hat{U}$ on $\hat{\rho}_i$ which rotates the Bloch vector to $\mathbf{r}'_i$ and brings it to a new state $\hat{\rho}'_i$ diagonal in the energy eigenbasis with the eigenvalues in ascending order. The thermalization dynamics of both states are pictured on the Bloch sphere in Fig. 1(a). The color bar indicates the nonequilibrium free energy F_{neq} , which is shown also in panel (b). The crossing of the two lines indicates a genuine quantum Mpemba effect. The inset shows the L_1 distance from the steady state $D_{L_1}[\hat{\rho}(t), T_b] = \sum_{i,j} |\hat{\rho}^{ij}(t) - \hat{\tau}_\beta^{ij}|$, that was studied also in the first experiment on classical Markovian systems [16] and indicates that the thermalization for the transformed state $\hat{\rho}'_i$ is exponentially accelerated. We now study the division of the entropy production defined by Eq. (7) in Fig. 1(c). It can be seen that for the transformed state $\hat{\rho}'_i$ the coherent contribution is zero since the overlap with the coherent modes has been eliminated. The spectrum of the Davies map is shown as an inset. Note that for a single qubit, at any temperature T_b the spectral gap is defined by a complex eigenpair.

Example 2: Many qubits—We now shift our focus to a more complex system and consider a transverse-field Ising model (TFIM) with open boundary conditions $\hat{H} = -J \sum_{j=1}^{L-1} \hat{\sigma}_j^z \hat{\sigma}_{j+1}^z + h \sum_{j=1}^L \hat{\sigma}_j^x$. We construct a random mixed state $\hat{\rho}_i$ by averaging over 1000 random pure states. Then, as for the single qubit case, we obtain the transformed state $\hat{\rho}'_i$ by applying $\hat{U}$. In panel (a) of Fig. 2 we compute the irreversible entropy production. The main difference with the single qubit case is that since the overlap of the random state $\hat{\rho}_i$ with the slowest mode $\hat{\lambda}_2$ is small, the difference with the transformed state manifests only at a later time. The situation changes when one considers an initial state $\hat{\rho}_i$ which has a larger overlap with $\hat{\lambda}_2$. We construct such a state by applying a Metropolis-based transformation, which is explained in the Supplemental

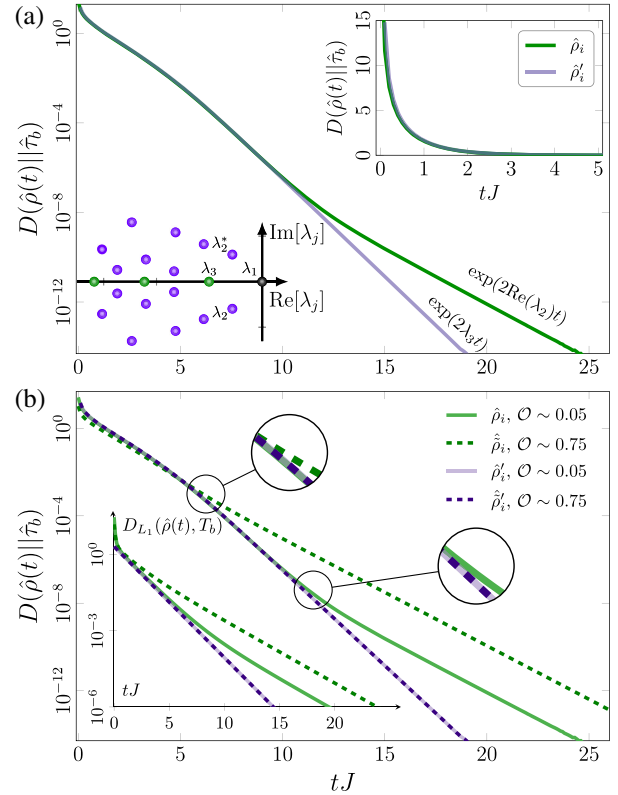


FIG. 2. The genuine quantum Mpemba effect in a many-qubit system manifests at different timescales. (a) We study the total entropy production D for the transverse field Ising model with anisotropy at $h = J/2$ with $L = 5$ spins and bath temperature $T_b = 0.1J$. The rotation $\hat{U}$ allows for an exponential speedup governed by λ_3 that shows up at timescales much larger than for the single qubit case. The upper inset shows D in linear scale while the lower inset displays the first 20 eigenvalues of the spectrum of the Davies generator $\mathcal{G}$. (b) An initial state with a large overlap with the two slowest decaying modes $\mathcal{O} = |\text{Tr}(\hat{\lambda}_2^* \hat{\rho}_i)| + |\text{Tr}(\hat{\lambda}_2 \hat{\rho}_i)|$ achieves the speedup sooner than one with a small overlap. For $\mathcal{O} \sim 0.05$ (full green line), the curves for the total entropy production cross around $t/J \sim 10$, while with $\mathcal{O} \sim 0.75$ (dashed green line), they coalesce at half of this time.

Material [50]. The dashed green line in panel (b) shows that similarly to the single qubit case, the exponential difference in the thermalization speed of $\hat{\rho}_i$ and $\hat{\rho}'_i$ is evident already at short times.

Physical realizations—For many physically relevant scenarios, starting from the microscopic Hamiltonian for system and environment and following the standard steps [63], one can derive an effective Davies map description. These include, for instance, quantum optics setups [64] and mesoscopic heat engines [65]. Importantly, Davies maps can also be used to accurately estimate the impact of noise on single-qubit and two-qubit operations in superconducting transmon architectures [66]. For concreteness, in Sec. D of the Supplemental Material [50], we consider the Davies map description of a spinful fermionic quantum dot

coupled to a heat bath and a two-level atom in a photonic environment and reveal the occurrence of the Mpemba effect for experimental parameters. Note also that the classical and quantum contributions to the entropy production [Eq. (7)] have recently been measured in a table-top optical experiment described by a Davies map [67].

Conclusion—We have demonstrated that a quantum Mpemba effect can be engineered by exploiting the sub-block structure of the generator in thermalizing open Markovian systems. We highlight the nonequilibrium free energy as a central thermodynamic object to both define and study the effect in open quantum systems. When the spectral gap of a Davies generator is defined by a complex eigenpair a unitary operation can always be found on an initial state that simultaneously raises the free energy and provides exponentially quicker thermalization. The work presented here not only enhances our understanding of the effect in quantum systems and should inspire further studies from the perspective of quantum thermodynamics, but it helps bridge the gap between quantum and classical systems and could prove useful in dissipative state engineering protocols [68–70].

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